Solution Requirements

Date	30 June 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:
Identify and document the requirements of the OptiCrop system to clearly specify its functionalities and performance expectations. Functional Requirements describe the features and services that the system must provide, while Non-Functional Requirements define the quality attributes, constraints, and performance standards that ensure the system operates efficiently, reliably, and securely.

Functional Requirements (FR): Define the core features and operations of the OptiCrop application, such as crop recommendation, input validation, prediction generation, and result visualization.
Non-Functional Requirements (NFR): Define the performance, scalability, security, reliability, usability, and maintainability requirements that the OptiCrop system must satisfy to deliver a high-quality user experience.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The application can be used without creating an account. In future versions, secure user authentication may be added to enable personalized recommendations and user history.	Medium

2	Authorization levels	The system may provide different user roles such as Farmers, Agricultural Experts, Researchers, and Administrators with role-based access to system features.	Medium
3	External interfaces	The application communicates with the trained machine learning model and crop dataset, with the possibility of integrating weather services and agricultural APIs in future releases.	High
4	Transactions processing	The system receives soil and climate parameters, validates the entered values, processes them through the prediction model, and returns the recommended crop instantly.	High
5	Reporting	The application presents the recommended crop, prediction confidence (optional), alternative crop options, and graphical summaries where applicable.	High
6	Business rules, etc.	Recommendations are generated only when valid input values are provided. All predictions follow the trained ML model and predefined business rules.	High

7	Compliance to laws or regulations	The application should follow responsible AI practices and ensure secure handling of user information while complying with applicable privacy guidelines.	Medium
8	*Other*	The system should provide a responsive interface and allow future expansion with modules such as fertilizer recommendation, disease detection, and yield prediction.	Medium

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	The system should process user inputs efficiently and generate recommendations with minimal delay..	Prediction results displayed in **less than 2 seconds**.
2	Scalability	The application should support growth in users, datasets, and additional functionalities while maintaining consistent performance.	Supports **500+ concurrent users** with consistent response times.
3	Security & Data Privacy	User inputs and application data should be protected from unauthorized access and misuse.	Secure data handling with **input validation and HTTPS support**.

4	Reliability & Availability	The application should remain operational and accessible with minimal downtime.	**99% uptime** and stable prediction service availability.
5	Usability & Accessibility	The interface should be simple, responsive, and easy to use for farmers and agricultural stakeholders.	Responsive design supporting **desktop, tablet, and mobile devices**.
6	Other	The system should be easy to maintain, update, and deploy across different environments.	Compatible with **Windows, Linux, and cloud deployment platforms** such as Render.